Tamanjyot Singh

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The University of Texas at Dallas, Richardson, TX, USA

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Research Interests

Relativistic Astrophysics | Black Hole Spin | Tidal Disruption Events | Gravitational Waves (GWs)
Precession in Binary Black Holes | Gravitational Lensing of GWs

Education

Ph.D. **The University of Texas at Dallas** — Richardson, TX, USA
Physics — January 2021 - Ongoing *(expected April 2026)*
Thesis: *The Black Hole Ballet*: Spin and precession effects on gravitational waves and tidal disruption events
Advisor: Dr. Michael Kesden

M.S. **The University of Texas at Dallas** — Richardson, TX, USA
Physics — January 2021 - December 2023
GPA: 3.93/4.0

B.Tech. **Punjab Engineering College (Deemed to be University)** — Chandigarh, India
Aerospace Engineering with minor in Mathematics — August 2016 - June 2020
GPA: 9.2/10.0, Rank 2
Major project: Computational study of flame propagation in a micro-scale channel and applications of micro-combustors in thermo-electric power generation.

Experience

Research assistant — Richardson, TX, USA
Department of Physics, The University of Texas at Dallas — May 2022 - Ongoing

- **Precession and Nutation in gravitational waves**: Investigating spin-orbit and spin-spin coupling in binary black hole systems and their gravitational wave signals.
- **Lensing vs Precession in gravitational waves**: Distinguishing between a lensed and a precessing gravitational wave signal.
- **Tidal disruption events**: Exploring inclination biases due to the spin of the supermassive black hole in stellar disruptions.

Teaching assistant — Richardson, TX, USA
Department of Physics, The University of Texas at Dallas — January 2021 - May 2022

- **Physics Laboratory I - Mechanics**: Introductory physics laboratory course.

Research Intern — Mumbai, India
Combustion Research Laboratory, Indian Institute of Technology, Bombay — January 2019 - April 2019

- **Combustion characteristics and effect of hydrogen content on syngas-air mixtures**: Formulated a machine learning based empirical model to predict laminar burning velocities of syngas-air mixtures. Contributed to: Varghese and Kumar 2020

Indian Air Force (IAF) Intern — Chandigarh, India
3rd Base Repair Depot (3BRD), Air Force Station — June 2018

Publications

Published:

Tamanjyot Singh, Michael Kesden — February 2024

"Distribution of orbital inclinations for tidal disruption events by Kerr black holes"

Phys. Rev. D 109, 043016, ADS

Submitted:

Tamanjyot Singh, Evangelos Stoikos, Saif Ali, Nathan Steinle, Michael Kesden, and Lindsay King — September 2025

"Detecting regular precession using a new gravitational waveform model directly parameterized by both precession amplitude and frequency"

Submitted to Phys. Rev. D, arXiv: 2509.10628, ADS

In preparation:

Tamanjyot Singh, Michael Kesden, and Lindsay King

"Detecting nutation: Generic precession in binary black holes"

Tien Nguyen, ***Tamanjyot Singh***, Benjamin McCallip, Michael Kesden, and Lindsay King

"Distinguishing lensing and precession in binary black holes"

Research Presentations

Contributed	**Team for EXtragalactic AStrophysics (TEXAS) – Partnerships in Astronomy & Astrophysics Research and Education (PAARE) Annual Symposium 2025**, The University of Texas at Dallas, Richardson, TX	April 2025
Contributed	**APS Global Physics Summit 2025**, Anaheim, CA	March 2025
Contributed	**Fall meeting 2024, Texas Section APS**, Southern Methodist University, Dallas, TX	October 2024
Invited Visit	Center for Computational Astrophysics, Flatiron Institute, New York City, NY	May 2024
Contributed	**APS April meeting 2024**, Sacramento, CA	April 2024
Contributed	**Team for EXtragalactic AStrophysics (TEXAS) – Partnerships in Astronomy & Astrophysics Research and Education (PAARE) Annual Symposium 2024**, University of North Texas, Denton, TX	March 2024
Contributed	**10th Gulf Coast Gravity Meeting 2024**, University of Texas, Austin, TX	March 2024
Contributed	**International Conference on Gravitation and Cosmology 2023**, Indian Institute of Technology, Guwahati, Assam, India	December 2023
Contributed	**Fall meeting 2023, Texas Section APS**, Angelo State University, San Angelo, TX	October 2023
Contributed	**Spring meeting 2023, Texas Section APS**, Texas A&M University, Commerce, TX	March 2023

Recognitions

NSM Conference Travel Award — March 2025
School of Natural Sciences and Mathematics, The University of Texas at Dallas, TX

APS April meeting DGRAV Travel Award — April 2024
Division of Gravitational Physics, APS

Margie Renfrow travel award — March 2024
Department of Physics, The University of Texas at Dallas, TX

Teaching Assistant award 2022 — December 2022
Department of Physics, The University of Texas at Dallas, TX

PECOSA Sydney Prize (Awarded to 3 out of 700+ students) — September 2018
Punjab Engineering College, Chandigarh

1st position in the Mathematics Quiz — September 2016
Punjab State Council for Science and Technology, Chandigarh

99.3%ile in JEE (Mains) 2016 (among 1,500,000+ candidates) — April 2016

International Rank 76 in International Olympiad of Science 2013 — November 2013
(Gold Medal + *Special Achievement Certificate*)

Mentoring

I have had the pleasure of mentoring with many undergraduate and near-peer mentees over the years:

Austin Crawford	UTD grad student Fall 2024 - Ongoing *(Grad Student)*
Benjamin McCallip	Summer REU student 2025 *(Undergrad student)*
Tien Nguyen	PAARE fellow 2023-24 *(Post-baccalaureate student)*
Maya Basu	Summer REU student 2024 *(Undergrad Student)*
Tinatin Tsiskaridze	Summer exchange student 2024 *(Grad Student)* (from Tblisi State University, Georgia)
Niel Shah	Summer REU student 2023 *(Undergrad Student)*
Deven Linthicum	Summer REU student 2023 *(Undergrad Student)*
Allison Powell	Summer REU student 2022 *(Undergrad Student)*

Service and Leadership

Referee — Physical reviews D, Monthly Notices of the Royal Astronomical Society

Journal Club host: Hosting and leading weekly journal club meetings with astrophysics students and faculty across DFW in a joint effort between the University of Texas at Dallas, University of North Texas, Denton, and University of Texas at Arlington. — August 2024 - August 2025

Course Facilitator: Volunteered to teach and translate Punjabi along with hosting biweekly community sessions, fostering an environment for positive outlook on life as Sikhs in the USA. — August 2024 - May 2025

Graduate Students in Physics - Vice President: Representing graduate student interests in the department. Assisting President in maintenance and operations of the organization. — May 2023 - July 2024
Notable: Procured a department-wide Overleaf license.

Graduate Students in Physics - Astronomy committee head: Graduate student liaison for all things telescope, hosting star-watch and lunar eclipse parties. Responsible for maintenance of the existing equipment and inventory along with buying new equipment. — March 2022 - February 2024

Technical skills

- Python: `numpy`, `matplotlib`, `scipy`, `PyCBC`, `astropy`, etc.
- Other programming languages: C, C++, Matlab
- General: `git`, LaTeX, `HTML`, cluster computing (`slurm`)

Professional Development

Participant in **Prospects in Theoretical Physics 2025** — July 2025
Institute for Advanced Study, Princeton, NJ

Participant in **Entering Mentoring** — July 2024
The University of Texas at Dallas, Richardson, TX

Course Certificate for **Astro 101 : Black Holes** — June 2020
University of Alberta on Coursera.

Host for workshop on **Gravitational lensing** (Undergraduate) — October 2019
Punjab Engineering College, Chandigarh

Participant in **ASTROWIN**, Winter School on Astronomy 2019 — February 2019
Birla Museum, Hyderabad

Course Certificate for **Particle Physics: an Introduction** — July 2018
University of Geneva on Coursera.

Course Certificate for **Understanding Einstein: The Special Theory of Relativity** — July 2017
Stanford University on Coursera.